JEE Main 2027 — Model Question Paper 1

Duration: 3 hours · Maximum marks: 300 · 75 questions (25 each in Physics, Chemistry, Mathematics)

1. Section A of each subject: 20 multiple-choice questions, one correct option. +4 for correct, −1 for wrong, 0 if unattempted.
2. Section B of each subject: 5 numerical-value questions; enter the exact integer answer. +4 for correct, 0 otherwise (no negative marking).
3. Take g = 10 m/s² unless stated otherwise.
4. Answer key is at the end of the paper.

Physics

Section A — Multiple Choice (single correct)

1. A particle moves along the x-axis as $x = 4t^2 - t^3$ (x in m, t in s). Its velocity at the instant its acceleration becomes zero is

(1) 8/3 m/s **(2)** 4 m/s
(3) 16/3 m/s **(4)** 32/3 m/s

2. A ball projected at 45° to the horizontal has a horizontal range of 40 m. The maximum height reached is

(1) 20 m **(2)** 10 m
(3) 5 m **(4)** 40 m

3. A block is released from rest on a rough incline of angle 37° (μ = 0.5, g = 10 m/s², sin 37° = 0.6). Its acceleration down the incline is

(1) 2 m/s² **(2)** 4 m/s²
(3) 6 m/s² **(4)** 1 m/s²

4. A mass m is attached to two identical springs of constant k. The ratio of time periods when the springs are connected in series to when they are in parallel is

(1) 1 : 2 **(2)** 4 : 1
(3) 1 : 1 **(4)** 2 : 1

5. For a solid sphere rolling without slipping, the ratio of rotational kinetic energy to total kinetic energy is

(1) 2/5 **(2)** 2/7
(3) 5/7 **(4)** 1/2

6. A satellite orbits the Earth at radius 4R (R = radius of Earth). If v_0 is the orbital speed of a satellite just above the surface, its orbital speed is

(1) $v_0/2$ **(2)** $v_0/4$
(3) $2v_0$ **(4)** $v_0/\sqrt{2}$

7. A planet has twice the mass and twice the radius of Earth. Escape speed from it is (v_e = 11.2 km/s for Earth)

(1) 22.4 km/s **(2)** 5.6 km/s
(3) 15.8 km/s **(4)** 11.2 km/s

8. A wire extends by x under a load. Another wire of the same material with double the length and double the diameter, under the same load, extends by

(1) x **(2)** 2x
(3) x/2 **(4)** x/4

9. A monatomic ideal gas at 300 K is compressed adiabatically to 1/8 of its volume. Its final temperature is

(1) 600 K **(2)** 1200 K
(3) 2400 K **(4)** 900 K

10. A Carnot engine with sink at 300 K has efficiency 25%. To raise the efficiency to 50% keeping the sink fixed, the source temperature must be increased by

(1) 200 K **(2)** 100 K
(3) 300 K **(4)** 400 K

11. The rms speed of oxygen molecules equals that of hydrogen molecules at 300 K when the oxygen is at

(1) 1200 K **(2)** 2400 K
(3) 4800 K **(4)** 9600 K

12. For a uniformly charged ring of radius R, the electric field on its axis is maximum at a distance from the centre of

(1) R **(2)** R/2
(3) $\sqrt{2}$ R **(4)** R/$\sqrt{2}$

13. A parallel plate capacitor C has half its gap (thickness d/2, full area) filled with a dielectric of K = 3. The new capacitance is

(1) 1.5 C **(2)** 3 C
(3) 2 C **(4)** 0.75 C

14. Three 6 Ω resistors in parallel are connected in series with a 2 Ω resistor across an ideal 12 V battery. Current drawn from the battery is

(1) 2 A **(2)** 1.5 A
(3) 3 A **(4)** 6 A

15. A wire bent into a semicircle of radius R carries current I. The magnetic field at its centre (ignoring straight leads) is

(1) $\mu_0 I/2R$ **(2)** $\mu_0 I/4R$
(3) $\mu_0 I/4\pi R$ **(4)** $\mu_0 I/\pi R$

16. A proton and an α-particle having equal kinetic energies enter a uniform magnetic field perpendicularly. The ratio of their radii $r_\alpha : r_p$ is

(1) 1 : 1 **(2)** 2 : 1
(3) 1 : 2 **(4)** $\sqrt{2}$: 1

17. In a series LCR circuit, L = 0.1 H and C = 10 μF. The resonant angular frequency is

(1) 100 rad/s **(2)** 10^4 rad/s
(3) 316 rad/s **(4)** 1000 rad/s

18. Light of photon energy 4 eV falls on a metal of work function 2 eV. If the intensity is doubled, the stopping potential becomes

(1) 4 V **(2)** 1 V
(3) 2 V **(4)** 0.5 V

19. The de Broglie wavelength of an electron accelerated from rest through 100 V is about

(1) 12.3 Å **(2)** 1.23 Å

(3) 0.123 Å **(4)** 123 Å

20. A radioactive sample has half-life 10 days. The percentage that has decayed after 30 days is

(1) 87.5% **(2)** 75%

(3) 12.5% **(4)** 66.7%

Section B — Numerical Value

21. A convex lens of focal length 20 cm forms a real image of an object kept 30 cm from it. The image distance (in cm) is

Answer: _____

22. Electrons in a sample of hydrogen atoms are excited to n = 4. The maximum number of distinct spectral lines emitted is

Answer: _____

23. The force between point charges 2 μC and 3 μC placed 30 cm apart in vacuum is F. F in millinewton is

Answer: _____

24. A 2 kg body moving at 10 m/s collides with a stationary 3 kg body and they stick together. Kinetic energy lost (in J) is

Answer: _____

25. In a metre bridge, the null point is at 40 cm from the left end with an unknown resistance X in the left gap and 30 Ω in the right gap. X (in Ω) is

Answer: _____

Chemistry

Section A — Multiple Choice (single correct)

26. The number of moles in 4.4 g of CO_2 is

(1) 1 **(2)** 0.1

(3) 0.01 **(4)** 0.44

27. Which has the highest first ionisation enthalpy?

(1) O **(2)** C

(3) N **(4)** B

28. The shape of XeF_4 is

(1) tetrahedral **(2)** see-saw

(3) square pyramidal **(4)** square planar

29. The bond order of O_2^+ is

(1) 2.5 **(2)** 2

(3) 1.5 **(4)** 3

30. Hybridisation of S in SF_6 is

(1) sp^3d **(2)** sp^3d^2

(3) sp^3 **(4)** dsp^2

31. pH of 0.01 M NaOH at 25 °C is

(1) 2 **(2)** 10

(3) 12 **(4)** 13

32. For $N_2 + 3H_2 \rightleftharpoons 2NH_3$, $Kp = Kc(RT)^x$ where x is

(1) -2 **(2)** 2

(3) -1 **(4)** 0

33. Which of the following is aromatic?

(1) cyclopentadienyl cation **(2)** cyclooctatetraene

(3) cyclopentadiene **(4)** cyclopentadienyl anion

34. Major product of addition of HBr to propene in the presence of peroxide is

(1) 2-bromopropane **(2)** 1-bromopropane

(3) 1,2-dibromopropane **(4)** propan-2-ol

35. The strongest acid among the following is

(1) phenol **(2)** p-cresol

(3) p-nitrophenol **(4)** p-methoxyphenol

36. Which alcohol gives turbidity with Lucas reagent immediately?

(1) 2-methylpropan-2-ol **(2)** butan-1-ol

(3) butan-2-ol **(4)** methanol

37. The ground state electronic configuration of Cr (Z = 24) is

(1) $[Ar]\ 3d^4\ 4s^2$ **(2)** $[Ar]\ 3d^6$

(3) $[Ar]\ 3d^5\ 4s^1$ **(4)** $[Ar]\ 3d^3\ 4s^2\ 4p^1$

38. Spin-only magnetic moment of Fe^{3+} is

(1) 4.90 BM **(2)** 5.92 BM

(3) 3.87 BM **(4)** 1.73 BM

39. The oxidation state of Cr in $K_2Cr_2O_7$ is

(1) +3 **(2)** +7

(3) +4 **(4)** +6

40. As a consequence of lanthanoid contraction, the pair with almost identical atomic radii is

(1) Zr and Hf **(2)** Ti and Zr

(3) Sc and Y **(4)** Cu and Ag

41. Ozonolysis of but-2-ene followed by Zn/H_2O gives

(1) propanal + methanal **(2)** ethanal only

(3) propanone **(4)** butanal

42. $[Co(NH_3)_6]^{3+}$ is

(1) outer orbital, paramagnetic **(2)** outer orbital, diamagnetic

(3) inner orbital, diamagnetic **(4)** inner orbital, paramagnetic

43. The number of geometrical isomers of square planar $[Pt(NH_3)_2Cl_2]$ is

(1) 2 **(2)** 3

(3) 1 **(4)** 0

44. Which is a non-reducing sugar?

(1) glucose **(2)** maltose

(3) lactose **(4)** sucrose

45. The unit of rate constant of a first-order reaction is

(1) mol L^{-1} s^{-1} **(2)** s^{-1}

(3) L mol^{-1} s^{-1} **(4)** L^2 mol^{-2} s^{-1}

Section B — Numerical Value

46. The number of unpaired electrons in Mn^{2+} is

Answer: _____

47. 9.8 g of H_2SO_4 is dissolved in 500 g of water. The molality is $x \times 10^{-1}$ mol kg^{-1}. x is

Answer: _____

48. For the Daniell cell (n = 2), E° = 1.10 V. The magnitude of ΔG° in kJ mol^{-1} (nearest integer, F = 96500 C mol^{-1}) is

Answer: _____

49. The number of structurally isomeric alcohols with formula $C_4H_{10}O$ is

Answer: _____

50. A first-order reaction has a half-life of 20 min. Time (in min) for 75% completion is

Answer: _____

Mathematics

Section A — Multiple Choice (single correct)

51. If α, β are the roots of $x^2 - 5x + 6 = 0$, then $\alpha^3 + \beta^3$ equals

(1) 35 **(2)** 125

(3) 90 **(4)** 27

52. The number of distinct arrangements of the letters of LEVEL is

(1) 60 **(2)** 120

(3) 30 **(4)** 20

53. The coefficient of x^3 in $(1 + x)^{10}$ is

(1) 210 **(2)** 120

(3) 45 **(4)** 252

54. The value of $\lim_{x \to 0} \sin 3x / x$ is

(1) 1/3 **(2)** 0

(3) 1 **(4)** 3

55. If $y = x^x$, then dy/dx at x = 1 is

(1) 0 **(2)** e

(3) 1 **(4)** 2

56. $\int_0^{\pi/2} \sin^2 x \, dx$ equals

(1) π/4 **(2)** π/2

(3) 1/2 **(4)** π

57. The area enclosed between $y = x^2$ and y = x is

(1) 1/3 **(2)** 1/6

(3) 1/2 **(4)** 2/3

58. The solution of $dy/dx = y$ with $y(0) = 2$ is

(1) $y = e^x + 1$ **(2)** $y = 2x + 2$

(3) $y = e^{2x}$ **(4)** $y = 2e^x$

59. The distance of the point $(2, 3)$ from the line $3x + 4y - 3 = 0$ is

(1) 3 **(2)** 5

(3) 15 **(4)** 18/5

60. The eccentricity of the ellipse $x^2/25 + y^2/16 = 1$ is

(1) 4/5 **(2)** 5/3

(3) 3/5 **(4)** 16/25

61. The focus of the parabola $y^2 = 12x$ is

(1) $(0, 3)$ **(2)** $(3, 0)$

(3) $(12, 0)$ **(4)** $(-3, 0)$

62. If A is a 3×3 matrix with $|A| = 4$, then $|2A|$ is

(1) 8 **(2)** 16

(3) 64 **(4)** 32

63. If A is a 3×3 matrix with $|A| = 3$, then $|\text{adj } A|$ is

(1) 9 **(2)** 3

(3) 27 **(4)** 1/3

64. Two fair dice are thrown. The probability that the sum is 7 is

(1) 1/12 **(2)** 1/6

(3) 5/36 **(4)** 7/36

65. The variance of the first 10 natural numbers is

(1) 9.5 **(2)** 8.25

(3) 10 **(4)** 5.5

66. The principal value of $\sin^{-1}(\sin 5\pi/6)$ is

(1) $5\pi/6$ **(2)** $-\pi/6$

(3) $\pi/6$ **(4)** $\pi/3$

67. If $z = (1 + i)/(1 - i)$, then z^{2023} equals

(1) 1 **(2)** i

(3) -1 **(4)** $-i$

68. The angle between the vectors $\hat{i} + \hat{j}$ and $\hat{j} + \hat{k}$ is

(1) $\pi/3$ **(2)** $\pi/4$

(3) $\pi/6$ **(4)** $\pi/2$

69. The number of reflexive relations on a set with 3 elements is

(1) 512 **(2)** 8

(3) 64 **(4)** 27

70. The sum of the infinite geometric series $1 + 1/3 + 1/9 + \ldots$ is

(1) 3 **(2)** 3/2

(3) 4/3 **(4)** 2

Section B — Numerical Value

71. The number of distinct terms in the expansion of $(x + y + z)^4$ is

Answer: _____

72. The value of $1^2 + 2^2 + 3^2 + \ldots + 10^2$ is

Answer: _____

73. $\int_0^3 [x]\, dx$, where [x] is the greatest integer function, equals

Answer: _____

74. A line with equal positive intercepts a on both axes passes through (2, 3). The value of 2a is

Answer: _____

75. The number of real solutions of $x^2 - 3|x| + 2 = 0$ is

Answer: _____

Answer Key

Physics

1. (3)	**2.** (2)	**3.** (1)	**4.** (4)	**5.** (2)	**6.** (1)
7. (4)	**8.** (3)	**9.** (2)	**10.** (1)	**11.** (3)	**12.** (4)
13. (1)	**14.** (3)	**15.** (2)	**16.** (1)	**17.** (4)	**18.** (3)
19. (2)	**20.** (1)	**21.** 60	**22.** 6	**23.** 600	**24.** 60
25. 20					

Chemistry

26. (2)	**27.** (3)	**28.** (4)	**29.** (1)	**30.** (2)	**31.** (3)
32. (1)	**33.** (4)	**34.** (2)	**35.** (3)	**36.** (1)	**37.** (3)
38. (2)	**39.** (4)	**40.** (1)	**41.** (2)	**42.** (3)	**43.** (1)
44. (4)	**45.** (2)	**46.** 5	**47.** 2	**48.** 212	**49.** 4
50. 40					

Mathematics

51. (1)	**52.** (3)	**53.** (2)	**54.** (4)	**55.** (3)	**56.** (1)
57. (2)	**58.** (4)	**59.** (1)	**60.** (3)	**61.** (2)	**62.** (4)
63. (1)	**64.** (2)	**65.** (2)	**66.** (3)	**67.** (4)	**68.** (1)
69. (3)	**70.** (2)	**71.** 15	**72.** 385	**73.** 3	**74.** 10
75. 4					

Original model paper prepared for practice, following the latest announced exam pattern. Not an official NTA paper.